

Project 3: Apache Giraph

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Course: Social Network Analysis

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Problem 1: SimpleShortestPaths Algorithm

Executed the pre-implemented SimpleShortestPaths algorithm on the provided `tiny_graph.txt` file in HDFS.

Command:

```
hadoop jar
giraph-examples-1.1.0-for-hadoop-1.2.1-jar-with-dependencies.jar
org.apache.giraph.GiraphRunner
org.apache.giraph.examples.SimpleShortestPathsComputation
-vif
org.apache.giraph.io.formats.JsonLongDoubleFloatDoubleVertexInputFormat
-vip /user/hduser/input/tiny_graph.txt
-vof org.apache.giraph.io.formats.IdWithValueTextOutputFormat
-op /user/hduser/output/shortestpaths -w 1
```

Results

Job ID: `job_202512220218_0001` **Task ID:** `task_202512220218_0001_m_000001`

Status: Completed Successfully

Output:

```
0      1.0
1      0.0
2      2.0
3      1.0
4      5.0
```

The results show shortest path distances from vertex 1 (source with distance 0.0) to all other vertices.

Problem 2: Label Propagation for Community Detection

Algorithm Overview

Implemented the Label Propagation algorithm for community detection:

1. **Initialization:** Each vertex starts with its own ID as its community label
2. **Iteration:** Each vertex adopts the majority label from its neighbors
3. **Tie-breaking:** If tied, adopt the smallest label value
4. **Termination:** Stops when no labels change or max iterations (50) reached

Implementation

Type Signature:

```
BasicComputation<LongWritable, DoubleWritable, FloatWritable, DoubleWritable>
```

Key Logic:

- Superstep 0: Initialize with vertex ID and send to all neighbors
- Supersteps 1-50: Count neighbor labels, adopt majority, propagate if changed
- Vote to halt when label stabilizes

Test Graph

Created a graph with two distinct communities:

```
Community 1: vertices {0, 1, 2} - fully connected triangle
Community 2: vertices {3, 4, 5} - fully connected triangle
Bridge: edge 2-3 connecting the communities
```

Input Format (JSON):

```
[0,0,[[1,1],[2,1]]]
[1,0,[[0,1],[2,1]]]
[2,0,[[0,1],[1,1],[3,1]]]
[3,0,[[2,1],[4,1],[5,1]]]
[4,0,[[3,1],[5,1]]]
[5,0,[[3,1],[4,1]]]
```

Execution

Build:

```
cd /home/sna/eclipse/workspace/giraph/giraph-examples
mvn clean package -DskipTests
```

Run:

```
hadoop jar
giraph-examples-1.1.0-for-hadoop-1.2.1-jar-with-dependencies.jar
org.apache.giraph.GiraphRunner
org.apache.giraph.examples.SimpleLabelPropagationComputation
-vif
org.apache.giraph.io.formats.JsonLongDoubleFloatDoubleVertexInputFormat
-vip /user/hduser/input/input_graph.txt
-vof org.apache.giraph.io.formats.IdWithValueTextOutputFormat
-op /user/hduser/output/labelpropagation -w 1
```

Results

Job ID: job_202512221340_0001 **Task ID:** task_202512221340_0001_m_000001

Status: Completed Successfully **Supersteps:** 5 **Vertices:** 6 **Edges:** 14

Community Detection Output:

Vertex	Community Label
0	0.0
1	0.0
2	0.0
3	2.0
4	2.0
5	2.0

Analysis:

- **Community 1** (label 0.0): Vertices {0, 1, 2}
- **Community 2** (label 2.0): Vertices {3, 4, 5}

The algorithm correctly identified the two communities. Vertex 2 remained in Community 1 because it has 2 connections to vertices 0,1 and only 1 connection to vertex 3 (majority voting).

Challenges

1. **Type Compatibility:** Initially used `LongWritable` for vertex values, but had to change to `DoubleWritable` to match the `JsonLongDoubleFloatDoubleVertexInputFormat` input format.
 2. **Maven Build Issues:** Encountered munge plugin errors, resolved by using `mvn clean` and fixing file ownership.
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Environment

- **VM:** Xubuntu 16.04 LTS
- **Hadoop:** 0.20.203.0
- **Giraph:** 1.1.0
- **Java:** Oracle Java 8

Conclusion

Successfully implemented and executed Label Propagation algorithm using Apache Giraph. The algorithm correctly detected communities in the test graph, demonstrating understanding of:

- Giraph's vertex-centric programming model
- Message passing between vertices
- Community detection through iterative label propagation